



Universal Definition of Myocardial Infarction

Kristian Thygesen, Joseph S. Alpert, Harvey D. White, on behalf of the Joint ESC/ACCF/AHA/WHF Task Force for the Redefinition of Myocardial Infarction
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ESC/ACCF/AHA/WHF EXPERT CONSENSUS DOCUMENT

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The recommendations set forth in this report are those of the Task Force Members and do not necessarily reflect the official position of the American College of Cardiology.

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Definition of Myocardial Infarction

Criteria for Acute Myocardial Infarction

The term myocardial infarction should be used when there is evidence of myocardial necrosis in a clinical setting consistent with myocardial ischaemia. Under these conditions any one of the following criteria meets the diagnosis for myocardial infarction:

- Detection of rise and/or fall of cardiac biomarkers (preferably troponin) with at least one value above the 99th percentile of the upper reference limit (URL) together with evidence of myocardial ischaemia with at least one of the following:
 - Symptoms of ischaemia;
 - ECG changes indicative of new ischaemia (new ST-T changes or new left bundle branch block [LBBB]);
 - Development of pathological Q waves in the ECG;
 - Imaging evidence of new loss of viable myocardium or new regional wall motion abnormality.
- Sudden, unexpected cardiac death, involving cardiac arrest, often with symptoms suggestive of myocardial ischaemia, and accompanied by presumably new ST elevation, or new LBBB, and/or evidence of fresh thrombus by coronary angiography and/or at autopsy, but death occurring before blood samples could be obtained, or at a time before the appearance of cardiac biomarkers in the blood.
- For percutaneous coronary interventions (PCI) in patients with normal baseline troponin values, elevations of cardiac biomarkers above the 99th percentile URL are indicative of peri-procedural myocardial necrosis. By convention, increases of biomarkers greater than $3 \times$ 99th percentile URL have been designated as defining PCI-related myocardial infarction. A subtype related to a documented stent thrombosis is recognized.
- For coronary artery bypass grafting (CABG) in patients with normal baseline troponin values, elevations of cardiac biomarkers above the 99th percentile URL are indicative of peri-procedural myocardial necrosis. By convention, increases of biomarkers greater than $5 \times$ 99th percentile URL plus either new pathological Q waves or new LBBB, or angiographically documented new graft or native coronary artery occlusion, or imaging evidence of new loss of viable myocardium have been designated as defining CABG-related myocardial infarction.
- Pathological findings of an acute myocardial infarction.

Criteria for Prior Myocardial Infarction

Any one of the following criteria meets the diagnosis for prior myocardial infarction:

- Development of new pathological Q waves with or without symptoms.
- Imaging evidence of a region of loss of viable myocardium that is thinned and fails to contract, in the absence of a non-ischaemic cause.
- Pathological findings of a healed or healing myocardial infarction.

Introduction

Myocardial infarction is a major cause of death and disability worldwide. Coronary atherosclerosis is a chronic disease with stable and unstable periods. During unstable periods with activated inflammation in the vascular wall, patients may develop a myocardial infarction. Myocardial infarction may be a minor event in a lifelong chronic disease, it may even go undetected, but it may also be a major catastrophic event leading to sudden death or severe haemodynamic deterioration. A myocardial infarction may be the first manifestation of coronary artery disease, or it may occur, repeatedly, in patients with established disease. Information on myocardial infarction attack rates can provide useful data regarding the burden of coronary artery disease within and across populations, especially if standardized data are col-

lected in a manner that demonstrates the distinction between incident and recurrent events. From the epidemiological point of view, the incidence of myocardial infarction in a population can be used as a proxy for the prevalence of coronary artery disease in that population. Furthermore, the term myocardial infarction has major psychological and legal implications for the individual and society. It is an indicator of one of the leading health problems in the world, and it is an outcome measure in clinical trials and observational studies. With these perspectives, myocardial infarction may be defined from a number of different clinical, electrocardiographic, biochemical, imaging, and pathological characteristics.

In the past, a general consensus existed for the clinical syndrome designated as myocardial infarction. In studies of disease prevalence, the World Health Organization (WHO) defined myocardial infarction from symptoms,

ECG abnormalities, and enzymes. However, the development of more sensitive and specific serological biomarkers and precise imaging techniques allows detection of ever smaller amounts of myocardial necrosis. Accordingly, current clinical practice, health care delivery systems, as well as epidemiology and clinical trials all require a more precise definition of myocardial infarction and a re-evaluation of previous definitions of this condition.

It should be appreciated that over the years, while more specific biomarkers of myocardial necrosis became available, the accuracy of detecting myocardial infarction has changed. Such changes occurred when glutamine-oxaloacetic transaminase (GOT) was replaced by lactate dehydrogenase (LDH) and later by creatine kinase (CK) and the MB fraction of CK, i.e. CKMB activity and CKMB mass. Current, more specific, and sensitive biomarkers and imaging methods to detect myocardial infarction are further refinements in this evolution.

In response to the issues posed by an alteration in our ability to identify myocardial infarction, the European Society of Cardiology (ESC) and the American College of Cardiology (ACC) convened a consensus conference in 1999 in order to re-examine jointly the definition of myocardial infarction (published in the year 2000 in the *European Heart Journal* and *Journal of the American College of Cardiology* [1]). The scientific and societal implications of an altered definition for myocardial infarction were examined from seven points of view: pathological, biochemical, electrocardiographic, imaging, clinical trials, epidemiological, and public policy. It became apparent from the deliberations of the former consensus committee that the term myocardial infarction should not be used without further qualifications, whether in clinical practice, in the description of patient cohorts, or in population studies. Such qualifications should refer to the amount of myocardial cell loss (infarct size), to the circumstances leading to the infarct (e.g. spontaneous or procedure related), and to the timing of the myocardial necrosis relative to the time of the observation (evolving, healing, or healed myocardial infarction) (1).

Following the 1999 ESC/ACC consensus conference, a group of cardiovascular epidemiologists met to address the specific needs of population surveillance. This international meeting, representing several national and international organizations, published recommendations in *Circulation* 2003 (2). These recommendations addressed the needs of researchers engaged in long-term population trend analysis in the context of changing diagnostic tools using retrospective medical record abstraction. Also considered was surveillance in developing countries and out-of-hospital death, both situations with limited and/or missing data. These recommendations continue to form the basis for epidemiological research.

Given the considerable advances in the diagnosis and management of myocardial infarction since the original document was published, the leadership of the ESC, the

ACC, and the American Heart Association (AHA) convened, together with the World Heart Federation (WHF), a Global Task Force to update the 2000 consensus document (1). As with the previous consensus committee, the Global Task Force was composed of a number of working groups in order to refine the ESC/ACC criteria for the diagnosis of myocardial infarction from various perspectives. With this goal in mind, the working groups were composed of experts within the field of biomarkers, ECG, imaging, interventions, clinical investigations, global perspectives, and implications. During several Task Force meetings, the recommendations of the working groups were coordinated, resulting in the present updated consensus document.

The Task Force recognizes that the definition of myocardial infarction will be subject to a variety of changes in the future as a result of scientific advance. Therefore, this document is not the final word on this issue for all time. Further refinement of the present definition will doubtless occur in the future.

Clinical Features of Ischaemia

The term myocardial infarction reflects cell death of cardiac myocytes caused by ischaemia, which is the result of a perfusion imbalance between supply and demand. Ischaemia in a clinical setting most often can be identified from the patient's history and from the ECG. Possible ischaemic symptoms include various combinations of chest, upper extremity, jaw, or epigastric discomfort with exertion or at rest. The discomfort associated with acute myocardial infarction usually lasts at least 20 min. Often, the discomfort is diffuse, not localized, not positional, not affected by movement of the region, and it may be accompanied by dyspnoea, diaphoresis, nausea, or syncope.

These symptoms are not specific to myocardial ischaemia and can be misdiagnosed and thus attributed to gastrointestinal, neurological, pulmonary, or musculoskeletal disorders. Myocardial infarction may occur with atypical symptoms, or even without symptoms, being detected only by ECG, biomarker elevations, or cardiac imaging.

Pathology

Myocardial infarction is defined by pathology as myocardial cell death due to prolonged ischaemia. Cell death is categorized pathologically as coagulation and/or contraction band necrosis, which usually evolves through oncosis, but can result to a lesser degree from apoptosis. Careful analysis of histological sections by an experienced observer is essential to distinguish these entities (1).

After the onset of myocardial ischaemia, cell death is not immediate but takes a finite period to develop (as little as 20 min or less in some animal models). It takes several hours before myocardial necrosis can be identified by macroscopic or microscopic post-mortem examination. Complete necrosis of all myocardial cells at risk requires at least 2–4 h or longer depending on the presence of collateral circulation to the ischaemic zone, persistent or intermittent coronary

arterial occlusion, the sensitivity of the myocytes to ischaemia, pre-conditioning, and/or, finally, individual demand for myocardial oxygen and nutrients. Myocardial infarctions are usually classified by size: microscopic (focal necrosis), small (<10% of the left ventricular [LV] myocardium), moderate (10–30% of the LV myocardium), and large (>30% of the LV myocardium), and by location. The pathological identification of myocardial necrosis is made without reference to morphological changes in the coronary arterial tree or to the clinical history (1).

Myocardial infarction can be defined pathologically as acute, healing, or healed. Acute myocardial infarction is characterized by the presence of polymorphonuclear leukocytes. If the time interval between the onset of the infarction and death is quite brief, e.g. 6 h, minimal or no polymorphonuclear leukocytes may be seen. The presence of mononuclear cells and fibroblasts, and the absence of polymorphonuclear leukocytes characterize healing infarction. Healed infarction is manifested as scar tissue without cellular infiltration. The entire process leading to a healed infarction usually takes at least 5–6 weeks. Reperfusion may alter the macroscopic and microscopic appearance of the necrotic zone by producing myocytes with contraction bands and large quantities of extravasated erythrocytes. Myocardial infarctions can be classified temporally from clinical and other features, as well as according to the pathological appearance, as evolving (<6 h), acute (6 h–7 days), healing (7–28 days), and healed (29 days and beyond). It should be emphasized that the clinical and electrocardiographic timing of the onset of an acute infarction may not correspond exactly with the pathological timing. For example, the ECG may still demonstrate evolving ST-T changes and cardiac biomarkers may still be elevated (implying a recent infarct) at a time when pathologically the infarction is in the healing phase (1).

Patients who suffer sudden cardiac death with or without ECG changes suggestive of ischaemia represent a challenging diagnostic group. Since these individuals die before pathological changes can develop in the myocardium, it is difficult to say with certainty whether these patients succumbed to a myocardial infarction or to an ischaemic event that led to a fatal arrhythmia. The mode of death in these cases is sudden, but the aetiology remains uncertain unless the individual reported previous symptoms of ischaemic heart disease prior to the cardiac arrest. Some patients with or without a history of coronary disease may develop clinical evidence of ischaemia, including prolonged and profound chest pain, diaphoresis and/or shortness of breath, and sudden collapse. These individuals may die before blood samples for biomarkers can be obtained, or these individuals may be in the lag phase before cardiac biomarkers can be identified in the blood. These patients may have suffered an evolving, fatal, acute myocardial infarction. If these patients present with presumably new ECG changes, for example ST elevation, and often with symptoms of ischaemia, they should be classified as having had a fatal myocardial infar-

Table 1. Clinical Classification of Different Types of Myocardial Infarction

Type 1	Spontaneous myocardial infarction related to ischaemia due to a primary coronary event such as plaque erosion and/or rupture, fissuring, or dissection
Type 2	Myocardial infarction secondary to ischaemia due to either increased oxygen demand or decreased supply, e.g. coronary artery spasm, coronary embolism, anaemia, arrhythmias, hypertension, or hypotension
Type 3	Sudden unexpected cardiac death, including cardiac arrest, often with symptoms suggestive of myocardial ischaemia, accompanied by presumably new STelevation, or new LBBB, or evidence of fresh thrombus in a coronary artery by angiography and/or at autopsy, but death occurring before blood samples could be obtained, or at a time before the appearance of cardiac biomarkers in the blood
Type 4a	Myocardial infarction associated with PCI
Type 4b	Myocardial infarction associated with stent thrombosis as documented by angiography or at autopsy
Type 5	Myocardial infarction associated with CABG

tion even if cardiac biomarker evidence of infarction is lacking. Also, patients with evidence of fresh thrombus by coronary angiography (if performed) and/or at autopsy should be classified as having undergone sudden death as a result of myocardial infarction.

Clinical Classification of Myocardial Infarction

Clinically the various types of myocardial infarction can be classified as shown in Table 1.

On occasion, patients may manifest more than one type of myocardial infarction simultaneously or sequentially. It should also be noted that the term myocardial infarction does not include myocardial cell death associated with mechanical injury from coronary artery bypass grafting (CABG), for example ventricular venting, or manipulation of the heart; nor does it include myocardial necrosis due to miscellaneous causes, e.g. renal failure, heart failure, cardioversion, electrophysiological ablation, sepsis, myocarditis, cardiac toxins, or infiltrative diseases.

Biomarker Evaluation

Myocardial cell death can be recognized by the appearance in the blood of different proteins released into the circulation from the damaged myocytes: myoglobin, cardiac troponin T and I, CK, LDH, as well as many others (3). Myocardial infarction is diagnosed when blood levels of sensitive and specific biomarkers such as cardiac troponin or CKMB are increased in the clinical setting of acute myocardial ischaemia (1). Although elevations in these biomarkers reflect myocardial necrosis, they do not indicate its mechanism (3,4). Thus, an elevated value of cardiac troponin in the absence of clinical evidence of ischaemia should

Table 2. Elevations of Troponin in the Absence of Overt Ischemic Heart Disease

Cardiac contusion, or other trauma including surgery, ablation, pacing, etc.
Congestive heart failure—acute and chronic
Aortic dissection
Aortic valve disease
Hypertrophic cardiomyopathy
Tachy- or bradyarrhythmias, or heart block
Apical ballooning syndrome
Rhabdomyolysis with cardiac injury
Pulmonary embolism, severe pulmonary hypertension
Renal failure
Acute neurological disease, including stroke or subarachnoid haemorrhage
Infiltrative diseases, e.g. amyloidosis, haemochromatosis, sarcoidosis, and scleroderma
Inflammatory diseases, e.g. myocarditis or myocardial extension of endo-/pericarditis
Drug toxicity or toxins
Critically ill patients, especially with respiratory failure or sepsis
Burns, especially if affecting >30% of body surface area
Extreme exertion

Modified from Jaffe *et al.* (4) and French and White (5).

prompt a search for other aetiologies of myocardial necrosis, such as myocarditis, aortic dissection, pulmonary embolism, congestive heart failure, renal failure, and other examples indicated in Table 2.

The preferred biomarker for myocardial necrosis is cardiac troponin (I or T), which has nearly absolute myocardial tissue specificity as well as high clinical sensitivity, thereby reflecting even microscopic zones of myocardial necrosis (3). An increased value for cardiac troponin is defined as a measurement exceeding the 99th percentile of a normal reference population (URL = upper reference limit). Detection of a rise and/or fall of the measurements is essential to the diagnosis of acute myocardial infarction (6). The above-mentioned discriminatory percentile is designated as the decision level for the diagnosis of myocardial infarction, and must be determined for each specific assay with appropriate quality control (7–9). Optimal precision (coefficient of variation [CV]) at the 99th percentile URL for each assay should be defined as $\leq 10\%$. Better precision ($CV \leq 10\%$) allows for more sensitive assays (10,11). The use of assays that do not have independent validation of optimal precision ($CV \leq 10\%$) is not recommended. The values for the 99th percentile can be found on the International Federation for Clinical Chemistry website http://www.ifcc.org/index.php?option=com_remository&Itemid=120&func=fileinfo&id=7.

Blood samples for the measurement of troponin should be drawn on first assessment (often some hours after the onset of symptoms) and 6–9 h later (12). An occasional patient may require an additional sample between 12 and 24 h if the earlier measurements were not elevated and the clinical suspicion of myocardial infarction is high (12). To establish the diagnosis of myocardial infarction, one elevated value above the decision level is required. The dem-

onstration of a rising and/or falling pattern is needed to distinguish background elevated troponin levels, e.g. patients with chronic renal failure (Table 2), from elevations in the same patients which are indicative of myocardial infarction (6). However, this pattern is not absolutely required to make the diagnosis of myocardial infarction if the patient presents >24 h after the onset of symptoms. Troponin values may remain elevated for 7–14 days following the onset of infarction (4).

If troponin assays are not available, the best alternative is CKMB (measured by mass assay). As with troponin, an increased CKMB value is defined as a measurement above the 99th percentile URL, which is designated as the decision level for the diagnosis of myocardial infarction (9). Gender-specific values should be employed (9). The CKMB measurements should be recorded at the time of the first assessment of the patient and 6–9 h later in order to demonstrate the rise and/or fall exceeding the 99th percentile URL for the diagnosis of myocardial infarction. An occasional patient may require an additional diagnostic sample between 12 and 24 h if the earlier CKMB measurements were not elevated and the clinical suspicion of myocardial infarction is high.

Measurement of total CK is not recommended for the diagnosis of myocardial infarction, because of the large skeletal muscle distribution and the lack of specificity of this enzyme.

Reinfarction

Traditionally, CKMB has been used to detect reinfarction. However, recent data suggest that troponin values provide similar information (13). In patients where recurrent myocardial infarction is suspected from clinical signs or symptoms following the initial infarction, an immediate measurement of the employed cardiac marker is recommended. A second sample should be obtained 3–6 h later. Recurrent infarction is diagnosed if there is a $\geq 20\%$ increase of the value in the second sample. Analytical values are considered to be different if they are different by >3 SDs of the variance of the measures (14). For troponin, this value is 5–7% for most assays at the levels involved with reinfarction. Thus, a 20% change should be considered significant, i.e. over that expected from analytical variability itself. This value should also exceed the 99th percentile URL.

Electrocardiographic Detection of Myocardial Infarction

The ECG is an integral part of the diagnostic work-up of patients with suspected myocardial infarction (1,2,15,16). The acute or evolving changes in the ST-T waveforms and the Q-waves when present potentially allow the clinician to date the event, to suggest the infarct-related artery, and to estimate the amount of myocardium at risk. Coronary artery dominance, size and distribution of arterial segments, col-

Table 3. ECG Manifestations of Acute Myocardial Ischaemia (in Absence of LVH and LBBB)

ST elevation
New ST elevation at the J-point in two contiguous leads with the cut-off points: ≥ 0.2 mV in men or ≥ 0.15 mV in women in leads V_2 – V_3 and/or ≥ 0.1 mV in other leads
ST depression and T-wave changes
New horizontal or down-sloping ST depression ≥ 0.05 mV in two contiguous leads; and/or T inversion ≥ 0.1 mV in two contiguous leads with prominent R-wave or R/S ratio >1

lateral vessels, and location, extent, and severity of coronary stenoses can also impact ECG manifestations of myocardial ischaemia (17). The ECG by itself is often insufficient to diagnose acute myocardial ischaemia or infarction since ST deviation may be observed in other conditions such as acute pericarditis, LV hypertrophy, LBBB, Brugada syndrome, and early repolarization patterns (18). Also Q-waves may occur due to myocardial fibrosis in the absence of coronary artery disease, as in, for example, cardiomyopathy.

ECG Abnormalities of Myocardial Ischaemia That May Evolve to Myocardial Infarction

ECG abnormalities of myocardial ischaemia or infarction may be inscribed in the PR segment, the QRS complex, and the ST segment or T-waves. The earliest manifestations of myocardial ischaemia are typical T-waves and ST segment changes (19,20). Increased hyper-acute T-wave amplitude with prominent symmetrical T-waves in at least two contiguous leads is an early sign that may precede the elevation of the ST segment. Increased R-wave amplitude and width (giant R-wave with S-wave diminution) are often seen in leads exhibiting ST elevation, and tall T-waves reflecting conduction delay in the ischaemic myocardium (21). Transient Q-waves may be observed during an episode of acute ischaemia or rarely during acute myocardial infarction with successful reperfusion (22).

Table 3 lists ECG criteria for the diagnosis of acute myocardial ischaemia that may lead to infarction. The J-point is used to determine the magnitude of the ST elevation. J-point elevation in men decreases with increasing age; however, that is not observed in women, in whom J-point elevation is less than in men (23).

Contiguous leads means lead groups such as anterior leads (V_1 – V_6), inferior leads (II, III, and aVF), or lateral/apical leads (I and aVL). More accurate spatial contiguity in the frontal plane can be established by the Cabrera display: aVL, I, aVR, II, aVF, and III (24). Supplemental leads such as V_3R and V_4R reflect the free wall of the right ventricle.

Although the criteria in Table 3 require that the ST shift be present in two or more contiguous leads, it should be noted that occasionally acute myocardial ischaemia may create sufficient ST segment shift to meet the criteria in one lead but have slightly less than the required ST shift in an adjacent contiguous lead. Lesser degrees of ST displacement or T-wave inversion in leads without prominent R-wave

amplitude do not exclude acute myocardial ischaemia or evolving myocardial infarction.

ST elevation or diagnostic Q-waves in regional lead groups are more specific than ST depression in localizing the site of myocardial ischaemia or necrosis (25,26). However, ST depression in leads V_1 – V_3 suggests myocardial ischaemia, especially when the terminal T-wave is positive (ST elevation equivalent), and may be confirmed by concomitant ST elevation ≥ 0.1 mV recorded in leads V_7 – V_9 (27,28). The term ‘posterior’ to reflect the basal part of the LV wall that lies on the diaphragm is no longer recommended. It is preferable to refer to this territory as infero-basal (29). In patients with inferior myocardial infarction it is advisable to record right precordial leads (V_3R and V_4R) seeking ST elevation in order to identify concomitant right ventricular infarction (30).

During an acute episode of chest discomfort, pseudo-normalization of previously inverted T-waves may indicate acute myocardial ischaemia. Pulmonary embolism, intracranial processes, or peri-/myocarditis may also result in ST-T abnormalities and should be considered (false positives) in the differential diagnosis.

The diagnosis of myocardial infarction is difficult in the presence of LBBB even when marked ST-T abnormalities or ST elevation are present that exceed standard criteria (31,32). A previous ECG may be helpful to determine the presence of acute myocardial infarction in this setting. In patients with right bundle branch block (RBBB), ST-T abnormalities in leads V_1 – V_3 are common, making it difficult to assess the presence of ischaemia in these leads; however, when ST elevation or Q-waves are found, myocardial ischaemia or infarction should be considered. Some patients present with ST elevation or new LBBB, and suffer sudden cardiac death before cardiac biomarkers become abnormal or pathological signs of myocardial necrosis become evident at autopsy. These patients should be classified as having had a fatal myocardial infarction.

Prior Myocardial Infarction

As shown in Table 4, Q-waves or QS complexes in the absence of QRS confounders are usually pathognomonic of a prior myocardial infarction (33–35). The specificity of the ECG diagnosis for myocardial infarction is greatest when Q-waves occur in several leads or lead groupings. ST deviations or T-waves alone are non-specific findings for

Table 4. ECG Changes Associated With Prior Myocardial Infarction

Any Q-wave in leads V_2 – V_3 ≥ 0.02 s or QS complex in leads V_2 and V_3
Q-wave ≥ 0.03 s and ≥ 0.1 mV deep or QS complex in leads I, II, aVL, aVF, or V_4 – V_6 in any two leads of a contiguous lead grouping (I, aVL, V_6 ; V_4 – V_6 ; II, III, and aVF)*
R-wave ≥ 0.04 s in V_1 – V_2 and R/S ≥ 1 with a concordant positive T-wave in the absence of a conduction defect

*The same criteria are used for supplemental leads V_7 – V_9 , and for the Cabrera frontal plane lead grouping.

myocardial necrosis. However, when these abnormalities occur in the same leads as the Q-waves, the likelihood of myocardial infarction is increased. For example, minor Q-waves ≥ 0.02 and < 0.03 s that are ≥ 0.1 mV deep are suggestive of prior infarction if accompanied by inverted T-waves in the same lead group.

Other validated myocardial infarction-coding algorithms, such as the Minnesota code, Novacode, and WHO MONICA, define Q-wave depth on the basis of depth, width, and ratio of R-wave amplitude, such as Q-wave depth at least one-third or one-fifth of R-wave amplitude, and have been used extensively in epidemiological studies and clinical trials (36,37).

Conditions That Confound the ECG Diagnosis of Myocardial Infarction

A QS complex in lead V₁ is normal. A Q-wave < 0.03 s and $< 1/4$ of the R-wave amplitude in lead III is normal if the frontal QRS axis is between 30 and 0°. The Q-wave may also be normal in aVL if the frontal QRS axis is between 60 and 90°. Septal Q-waves are small non-pathological Q-waves < 0.03 s and $< 1/4$ of the R-wave amplitude in leads I, aVL, aVF, and V₄-V₆. Pre-excitation, obstructive or dilated cardiomyopathy, LBBB, RBBB, left anterior hemiblock, left and right ventricular hypertrophy, myocarditis, acute cor pulmonale, or hyperkalaemia may be associated with Q/QS complexes in the absence of myocardial infarction. ECG abnormalities that simulate myocardial ischaemia or infarction are presented in Table 5.

Reinfarction

The ECG diagnosis of reinfarction following the initial infarction may be confounded by the initial evolutionary ECG changes. Reinfarction should be considered when ST elevation ≥ 0.1 mV reoccurs in a patient having a lesser degree of ST elevation or new pathognomonic Q waves, in at least two contiguous leads, particularly when associated

with ischaemic symptoms for 20 min or longer. The re-elevation of the ST segment can, however, also be seen in threatening myocardial rupture and should lead to additional diagnostic work-up. ST depression or LBBB on their own should not be considered valid criteria for myocardial infarction.

Coronary Revascularization

ECG abnormalities during or after percutaneous coronary intervention (PCI) are similar to those seen during spontaneous myocardial infarction. In patients who have undergone CABG, new ST-T abnormalities are common but not necessarily diagnostic of myocardial ischaemia (38). However, when new pathological Q waves (Table 4) appear in territories other than those identified before surgery, myocardial infarction should be considered, particularly if associated with elevated biomarkers, new wall motion abnormalities, or haemodynamic instability.

Imaging Techniques

Non-invasive imaging plays many roles in patients with known or suspected myocardial infarction, but this section concerns only its role in the diagnosis and characterization of infarction. The underlying rationale is that regional myocardial hypoperfusion and ischaemia lead to a cascade of events including myocardial dysfunction, cell death, and healing by fibrosis. Important imaging parameters are therefore perfusion, myocyte viability, myocardial thickness, thickening, and motion, and the effects of fibrosis on the kinetics of radiolabelled and paramagnetic contrast agents.

Commonly used imaging techniques in acute and chronic infarction are echocardiography, radionuclide ventriculography, myocardial perfusion scintigraphy (MPS), and magnetic resonance imaging (MRI). Positron emission tomography (PET) and X-ray computed tomography (CT) are less common. There is considerable overlap in their capabilities, but only the radionuclide techniques provide a direct assessment of myocardial viability because of the properties of the tracers used. Other techniques provide indirect assessments of myocardial viability, such as myocardial function from echocardiography or myocardial fibrosis from MRI.

Echocardiography

Echocardiography is an excellent real-time imaging technique with moderate spatial and temporal resolution. Its strength is the assessment of myocardial thickness, thickening, and motion at rest. This can be aided by tissue Doppler imaging. Echocardiographic contrast agents can improve endocardial visualization, but contrast studies are not yet fully validated for the detection of myocardial necrosis, although early work is encouraging (39).

Table 5. Common ECG Pitfalls in Diagnosing Myocardial Infarction

False positives
Benign early repolarization
LBBB
Pre-excitation
Brugada syndrome
Peri-/myocarditis
Pulmonary embolism
Subarachnoid haemorrhage
Metabolic disturbances such as hyperkalaemia
Failure to recognize normal limits for J-point displacement
Lead transposition or use of modified Mason-Likar configuration (24)
Cholecystitis
False negatives
Prior myocardial infarction with Q-waves and/or persistent ST elevation
Paced rhythm
LBBB

Radionuclide Imaging

Several radionuclide tracers allow viable myocytes to be imaged directly, including thallium-201, technetium-99m MIBI, tetrofosmin, and (^{18}F)2-fluorodeoxyglucose (FDG) (40–42). The strength of the techniques are that they are the only commonly available direct methods of assessing viability, although the relatively low resolution of the images disadvantages them for detecting small areas of infarction (43). The common single photon-emitting radio-pharmaceuticals are also tracers of myocardial perfusion and so the techniques readily detect areas of infarction and inducible per-fusion abnormalities. ECG-gated imaging provides a reliable assessment of myocardial motion, thickening, and global function (44,45).

Magnetic Resonance Imaging

Cardiovascular MRI has high spatial resolution and moderate temporal resolution. It is a well-validated standard for the assessment of myocardial function and has, in theory, similar capability to echocardiography in suspected acute infarction. It is, however, more cumbersome in an acute setting and is not commonly used. Paramagnetic contrast agents can be used to assess myocardial perfusion and the increase in extracellular space associated with the fibrosis of chronic infarction. The former is not yet fully validated in clinical practice, but the latter is well validated and can play an important role in the detection of infarction (46,47).

X-Ray Computed Tomography

Infarcted myocardium is initially visible to CT as a focal area of decreased LV enhancement, but later imaging shows hyper-enhancement as with late gadolinium imaging by MRI (48,49). This finding is clinically relevant because contrast-enhanced CT may be performed for suspected embolism and aortic dissection, conditions with clinical features that overlap with those of acute myocardial infarction.

Application in the Acute Phase of Myocardial Infarction

Imaging techniques can be useful in the diagnosis of myocardial infarction because of the ability to detect wall motion abnormalities in the presence of elevated cardiac bio-markers. If for some reason biomarkers have not been measured or may have normalized, demonstration of new loss of myocardial viability alone in the absence of non-ischaemic causes meets the criteria for myocardial infarction. However, if biomarkers have been measured at appropriate times and are normal, the determinations of these take precedence over the imaging criteria.

Echocardiography provides assessment of many non-ischaemic causes of acute chest pain such as perimyocarditis, valvular heart disease, cardiomyopathy, pulmonary embolism, or aortic dissection. Echocardiography is the imaging technique of choice for detecting complications of acute infarction including myocardial free wall rupture, acute ventricular septal defect, and mitral regurgitation

secondary to papillary muscle rupture or ischaemia. However, echocardiography cannot distinguish regional wall motion abnormalities due to myocardial ischaemia from infarction.

Radionuclide assessment of perfusion at the time of patient presentation can be performed with immediate tracer injection and imaging that can be delayed for up to several hours. The technique is interpreter dependent, although objective quantitative analysis is available. ECG gating provides simultaneous information on LV function.

An important role of acute echocardiography or radionuclide imaging is in patients with suspected myocardial infarction and a non-diagnostic ECG. A normal echocardiogram or resting ECG-gated scintigram has a 95–98% negative predictive value for excluding acute infarction (50–54). Thus, imaging techniques are useful for early triage and discharge of patients with suspected myocardial infarction (55,56).

A regional myocardial wall motion abnormality or loss of normal thickening may be caused by acute myocardial infarction or by one or more of several other ischaemic conditions including old infarction, acute ischaemia, stunning, or hibernation. Non-ischaemic conditions such as cardiomyopathy and inflammatory or infiltrative diseases can also lead to regional loss of viable myocardium or functional abnormality, and so the positive predictive value of imaging techniques is not high unless these conditions can be excluded and unless a new abnormality is detected or can be presumed to have arisen in the setting of other features of acute myocardial infarction.

Application in the Healing or Healed Phase of Myocardial Infarction

Imaging techniques are useful in myocardial infarction for analysis of LV function, both at rest and during dynamic exercise or pharmacological stress, to provide an assessment of remote inducible ischaemia. Echocardiography and radio-nuclide techniques, in conjunction with exercise or pharmacological stress, can identify ischaemia and myocardial viability. Non-invasive imaging techniques can diagnose healing or healed infarction by demonstrating regional wall motion, thinning, or scar in the absence of other causes.

The high resolution of contrast-enhanced MRI means that areas of late enhancement correlate well with areas of fibrosis and thereby enable differentiation between transmural and subendocardial scarring (57). The technique is therefore potentially valuable in assessing LV function and areas of viable and hence potentially hibernating myocardium.

Myocardial Infarction Associated With Revascularization Procedures

Peri-procedural myocardial infarction is different from spontaneous infarction, because the former is associated with the instrumentation of the heart that is required during

mechanical revascularization procedures by either PCI or CABG. Multiple events that can lead to myocardial necrosis are taking place, often in combination, during both types of intervention (58–61). While some loss of myocardial tissue may be unavoidable during procedures, it is likely that limitation of such damage is beneficial to the patient and their prognosis (62).

During PCI, myocardial necrosis may result from recognizable peri-procedural events, alone or in combination, such as side-branch occlusion, disruption of collateral flow, distal embolization, coronary dissection, slow flow or no-reflow phenomenon, and microvascular plugging. Embolization of intracoronary thrombus or atherosclerotic particulate debris cannot be entirely prevented despite current antithrombotic and antiplatelet adjunctive therapy or protection devices. Such events induce extensive inflammation of non-infarcted myocardium surrounding small islets of myocardium necrosis (63–67). New areas of myocardial necrosis have been demonstrated by MRI following PCI (68). A separate subcategory of myocardial infarction is related to stent thrombosis as documented by angiography and/or autopsy.

During CABG, numerous additional factors can lead to peri-procedural necrosis. These include direct myocardial trauma from sewing needles or manipulation of the heart, coronary dissection, global or regional ischaemia related to inadequate cardiac protection, microvascular events related to reperfusion, myocardial damage induced by oxygen free radical generation, or failure to reperfuse areas of the myocardium that are not subtended by graftable vessels (69–71). MRI studies suggest that most necrosis in this setting is not focal, but diffuse and localized to the sub-endocardium (72). Some clinicians and clinical investigators have preferred using CKMB for the diagnosis of peri-procedural infarction because of a substantial amount of data relating CKMB elevations to prognosis (73,74). However, an increasing number of studies using troponins in that respect have emerged (59,75).

Diagnostic Criteria for Myocardial Infarction With PCI

In the setting of PCI, the balloon inflation during a procedure almost always results in ischaemia whether or not accompanied by ST-T changes. The occurrence of procedure-related cell necrosis can be detected by measurement of cardiac biomarkers before or immediately after the procedure, and again at 6–12 and 18–24 h (76,77). Elevations of biomarkers above the 99th percentile URL after PCI, assuming a normal baseline troponin value, are indicative of post-procedural myocardial necrosis. There is currently no solid scientific basis for defining a biomarker threshold for the diagnosis of peri-procedural myocardial infarction. Pending further data, and by arbitrary convention, it is suggested to designate increases more than three times the 99th percentile URL as PCI-related myocardial infarction (type 4a).

If cardiac troponin is elevated before the procedure and not stable for at least two samples 6 h apart, there are insufficient data to recommend biomarker criteria for the diagnosis of peri-procedural myocardial infarction (77). If the values are stable or falling, criteria for reinfarction by further measurement of biomarkers together with the features of the ECG or imaging can be applied.

A separate subcategory of myocardial infarction (type 4b) is related to stent thrombosis as documented by angiography and/or autopsy. Although iatrogenic, myocardial infarction type 4b with verified stent thrombosis must meet the criteria for spontaneous myocardial infarction as well.

Diagnostic Criteria for Myocardial Infarction With CABG

Any increase of cardiac biomarkers after CABG indicates myocyte necrosis, implying that an increasing magnitude of biomarker is likely to be related to an impaired outcome. This has been demonstrated in clinical studies employing CKMB where elevations five, 10 and 20 times the upper limit of normal after CABG were associated with worsened prognosis (73,78,79). Likewise, the increase of troponin levels after CABG indicates necrosis of myocardial cells, which predicts a poor outcome, in particular when elevated to the highest quartile or quintile of the troponin measurements (59,75).

Unlike the prognosis, scant literature exists concerning the use of biomarkers for defining myocardial infarction in the setting of CABG. Therefore, biomarkers cannot stand alone in diagnosing myocardial infarction (type 5). In view of the adverse impact on survival observed in patients with significant biomarker elevations, this Task Force suggests, by arbitrary convention, that biomarker values more than five times the 99th percentile of the normal reference range during the first 72 h following CABG, when associated with the appearance of new pathological Q-waves or new LBBB, or angiographically documented new graft or native coronary artery occlusion, or imaging evidence of new loss of viable myocardium, should be considered as diagnostic of a CABG-related myocardial infarction (type 5 myocardial infarction).

Definition of Myocardial Infarction in Clinical Investigations

A universal definition for myocardial infarction would be of great benefit to future clinical studies in this area since it will allow for trial-to-trial comparisons as well as accurate meta-analyses involving multiple investigations. In clinical trials, myocardial infarction may be an entry criterion or an end-point. The definition of myocardial infarction employed in these trials will thus determine the characteristics of patients entering the studies as well as the number of outcome events. In recent investigations, different infarct definitions have been employed,

Table 6. Classification of the Different Types of Myocardial Infarction According to Multiples of the 99th Percentile URL of the Applied Cardiac Biomarker

Multiples × 99%	MI type 1 (spontaneous)	MI type 2 (secondary)	MI type 3* (sudden death)	MI type 4a† (PCI)	MI type 4b (stent thrombosis)	MI type 5† (CABG)	Total number
1–2 ×							
2–3 ×							
3–5 ×							
5–10 ×							
>10 ×							
Total number							

*Biomarkers are not available for this type of myocardial infarction since the patients expired before biomarker determination could be performed. †For the sake of completeness, the total distribution of biomarker values should be reported. The dark grey areas represent biomarker elevations below the decision limit used for these types of myocardial infarction. Irrespective of the specific end-point definition chosen in a clinical trial, all data should be provided. All boxes in the table should be completed, including the shaded areas.

thereby hampering comparison and generalization among these trials.

Consistency among investigators and regulatory authorities with regard to the definition of myocardial infarction used in clinical investigations is essential. The Task Force strongly encourages trialists to employ the definition described in this document. Furthermore, investigators should ensure that a trial provides comprehensive data for the various types of myocardial infarction (e.g. spontaneous, peri-procedural) and includes the employed decision limits for myocardial infarction of the cardiac biomarkers in question. All data should be made available to interested individuals in published format or on a website. Data concerning infarctions should be available in a form consistent with the current revised definitions of myocardial infarction. This does not necessarily restrict trialists to a narrow end-point definition, but rather ensures that across all future trials access to comparable data exists, thereby facilitating cross-study analyses. The recommendations put forward in this section are not detailed and should be supplemented by careful protocol planning and implementation in any clinical trial.

The Task Force strongly endorses the concept of the same decision limit for each biomarker employed for myocardial infarction types 1 and 2, and, likewise, the same higher three- and five-fold decision limits in the setting of myocardial infarction types 4a and 5, respectively (78–80) (Tables 6 and 7). In clinical trials, as in clinical practice, measurement of cardiac troponin T or I is preferred over

measurement of CKMB or other biomarkers for the diagnosis of myocardial infarction. Assessment of the quantity of myocardial damage (infarct size) is also an important trial end-point. Although the specific measurements vary depending on the assay and whether cardiac troponin T or I is used, in most studies troponin values correlate better with radionuclide- and MRI-determined infarct size than do CK and CKMB (81–83).

The use of cardiac troponins will undoubtedly increase the number of events recorded in a particular investigation because of increased sensitivity for detecting infarction (84–87). Ideally, data should be presented so that future clinical investigations or registries can translate the myocardial infarction end-point chosen in one study into the end-point of another study. Thus, measurements should be presented in a uniform manner to allow independent judgement and comparison of the clinical end-points. Furthermore, this Task Force suggests that data be reported as multiples of the 99th percentile URL of the applied biomarker, enabling comparisons between various classes and severity of the different types of myocardial infarction as indicated in Tables 6 and 7.

It is recommended that within a clinical trial all investigators whenever possible should employ the same assay in order to reduce the inter-assay variability, and, even better, the latter could be reduced to zero by application of a core laboratory using the same assay for all measurements.

In the design of a study, investigators should specify the expected effect of the new treatment under investigation. Factors that should be considered include:

- Assessment of the incidence of spontaneous myocardial infarction (type 1) and infarction related to myocardial oxygen supplies and demand (type 2) in treated patients vs. control subjects.
- Assessment of the incidence of sudden death related to myocardial infarction when applying the suggested criteria (type 3).
- Assessment of the incidence of procedure-related myocardial infarctions and biomarker elevations (PCI, types 4a and 4b; and CABG, type 5).

Table 7. Sample Clinical Trial Tabulation of Randomized Patients by Types of Myocardial Infarction

Types of MI	Treatment A Number of patients	Treatment B Number of patients
MI type 1		
MI type 2		
MI type 3		
MI type 4a		
MI type 4b		
MI type 5		
Total number		

Public Policy Implications of Redefinition of Myocardial Infarction

Evolution of the definition of a specific diagnosis such as myocardial infarction has a number of implications for individual citizens as well as for society at large. The process of assigning a specific diagnosis to a patient should be associated with a specific value for the patient. The resources spent on recording and tracking a particular diagnosis must also have a specific value to society in order to justify the effort. A tentative or final diagnosis is the basis for advice about further diagnostic testing, treatment, lifestyle changes, and prognosis for the patient. The aggregate of patients with a particular diagnosis is the basis for health care planning and policy, and resource allocation.

One of the goals of good clinical practice is to reach a definitive and specific diagnosis, which is supported by current scientific knowledge. The approach to the definition of myocardial infarction outlined in this document meets this goal. In general, the conceptual meaning of the term myocardial infarction has not changed, although new sensitive diagnostic methods have been developed to diagnose this entity. Thus, the current diagnosis of acute myocardial infarction is a clinical diagnosis based on patient symptoms, ECG changes, and highly sensitive biochemical markers, as well as information gleaned from various imaging techniques. However, it is important to characterize the extent of the infarct as well as residual LV function and the severity of coronary artery disease, rather than merely making a diagnosis of myocardial infarction. The information conveyed about the patient's prognosis and ability to work requires more than just the mere statement that the patient has suffered an infarct. The multiple other factors just mentioned are also required so that appropriate social, family, and employment decisions can be made. A number of risk scores have been developed predicting post-infarction prognosis. The classification of the various other prognostic entities associated with myocardial necrosis should lead to a reconsideration of the clinical coding entities currently employed for patients with the myriad conditions that can lead to myocardial necrosis with consequent elevation of biomarkers.

Many patients with myocardial infarction die suddenly. Difficulties in definition of sudden and out-of-hospital death make attribution of the cause of death variable among physicians, regions, and countries. For example, out-of-hospital death is generally ascribed to ischaemic heart disease in the USA but to stroke in Japan. These arbitrary and cultural criteria need re-examination.

It is important that any revised criteria for the definition of myocardial infarction involve comparability of this definition over time so that adequate trend data can be obtained. Furthermore, it is essential to ensure widespread availability and standard application of the measures in order to ensure comparability of data from various geo-

graphic regions. Shift in criteria resulting in a substantial increase or decrease in case identification will have significant health resource and cost implications (86,87). Moreover, an increase in sensitivity of the criteria for myocardial infarction might entail negative consequences for some patients who are not currently labelled as having had an infarction. On the other hand, increasing diagnostic sensitivity for myocardial infarction can have a positive impact for a society:

- Increasing the sensitivity of diagnostic criteria for myocardial infarction will result in more cases identified in a society, thereby allowing appropriate secondary prevention.
- Increasing the specificity of diagnostic criteria for myocardial infarction will result in more accurate diagnosis but will not exclude the presence of coronary artery disease, the cases of which may benefit from secondary prevention.

It should be appreciated that the agreed modification of the definition of myocardial infarction may be associated with consequences for the patients and their families with respect to psychological status, life insurance, professional career, as well as driving and pilot licences. Also the diagnosis is associated with societal implications as to diagnosis-related coding, hospital reimbursement, mortality statistics, sick leave, and disability attestation.

In order to meet this challenge, physicians must be adequately informed of the altered diagnostic criteria. Educational materials will need to be created and treatment guidelines must be appropriately adapted. Professional societies should take steps to facilitate the rapid dissemination of the revised definition to physicians, other health care professionals, administrators, and the general public.

Global Perspectives of the Redefinition of Myocardial Infarction

Cardiovascular disease is a global health problem. Approximately one-third of persons in the world die of cardiovascular disease, largely coronary artery disease and stroke, and 80% of these deaths from cardiovascular disease occur in developing countries. The greater proportion of deaths is due to heart disease and specifically coronary heart disease, of which myocardial infarction is a major manifestation. Since it is difficult to measure the prevalence of coronary artery disease in a population, the incidence of myocardial infarction may be used as a proxy, provided that a consistent definition is used when different populations, countries, or continents are being compared.

The changes in the definition of myocardial infarction have critical consequences for less developed and developing countries. In many countries, the resources to apply the new definition may not be available in all hospitals. However, many developing countries already do have medical facilities

capable of or currently employing the proposed definition of myocardial infarction. In the context of the overall cost for a patient with myocardial infarction, the expense associated with a troponin assay would not be excessive and should be economically affordable in many hospitals in developing countries, particularly those where infarcts are frequent events. The necessary equipment, staff training, and running costs may be lacking in some regions, but certainly not in others. In less advantaged hospitals, the diagnosis of myocardial infarction may depend mostly on clinical signs and symptoms coupled with less sophisticated biomarker analyses. Some of these institutions may only have access to CK and its isoenzymes at the present time. The redefinition arises from and is compatible with the latest scientific knowledge and with advances in technology, particularly with regard to the use of biomarkers, high quality electrocardiography, and imaging techniques. The definition can and should be used by developed countries immediately, and by developing countries as quickly as resources become available.

The change in the definition of myocardial infarction will have a substantial impact on the identification, prevention, and treatment of cardiovascular disease throughout the world. The new definition will impact epidemiological data from developing countries relating to prevalence and incidence. The simultaneous and continuing use of the older WHO definition for some years would allow a comparison between data obtained in the past and data to be acquired in the future employing the newer biomarker approach. Cultural, financial, structural, and organizational problems in the different countries of the world in diagnosis and therapy of acute myocardial infarction will require ongoing investigation. It is essential that the gap between therapeutic and diagnostic advances be addressed in this expanding area of cardiovascular disease.

Conflicts of Interest

The members of the Task Force established by the ESC, the ACCF, AHA and the WHF have participated independently in the preparation of this document, drawing on their academic and clinical experience and applying an objective and critical examination of all available literature. Most have undertaken and are undertaking work in collaboration with industry and governmental or private health providers (research studies, teaching conferences, consultation), but all believe such activities have not influenced their judgement. The best guarantee of their independence is in the quality of their past and current scientific work. However, to ensure openness, their relations with industry, government and private health providers are reported in Appendixes 1 and 2. Expenses for the Task Force/Writing Committee and preparation of this document were provided entirely by the above mentioned joint associations.

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REFERENCES

1. The Joint European Society of Cardiology/American College of Cardiology Committee. Myocardial infarction redefined—a consensus document of the Joint European Society of Cardiology/American College of Cardiology Committee for the Redefinition of Myocardial Infarction. *J Am Coll Cardiol* 2000;36:959–69.
2. Luepker RV, Apple FS, Christenson RH, et al. Case definitions for acute coronary heart disease in epidemiology and clinical research studies. A statement from the AHA Council on Epidemiology and Prevention; AHA Statistics Committee; World Heart Federation Council on Epidemiology and Prevention; the European Society of Cardiology Working Group on Epidemiology and Prevention; Centers for Disease Control and Prevention; and the National Heart, Lung, and Blood Institute. *Circulation* 2003;108:2543–9.
3. Jaffe AS, Ravkilde J, Roberts R, et al. It's time for a change to a troponin standard. *Circulation* 2000;102: 1216–20.
4. Jaffe AS, Babuin L, Apple FS. Biomarkers in acute cardiac disease. *J Am Coll Cardiol* 2006;48:1–11.
5. French JK, White HD. Clinical implications of the new definition of myocardial infarction. *Heart* 2004;90:99–106.
6. Jaffe AS. Chasing troponin: how low can you go if you can see the rise? *J Am Coll Cardiol* 2006;48:1763–4.
7. Panteghini M, Gerhardt W, Apple FS, Dati F, Ravkilde J, Wu AH. Quality specifications for cardiac troponin assays. *Clin Chem Lab Med* 2001;39:175–9.
8. Apple FS, Jesse RL, Newby LK, Wu AHB, Christenson RH. National Academy of Clinical Biochemistry and IFCC Committee for Standardization of Markers Cardiac Damage Laboratory Medicine Practice Guidelines: analytical issues for biochemical markers of acute coronary syndromes. *Circulation* 2007;115:e352–5.
9. Morrow DA, Cannon CP, Jesse RL, et al. National Academy of Clinical Biochemistry Laboratory Medicine Practice Guidelines: clinical characteristics and utilization of biochemical markers of acute coronary syndromes. *Circulation* 2007;115:e356–75.
10. Apple FS, Parvin CA, Buechler KF, Christenson RH, Wu AHB, Jaffe AS. Validation of the 99th percentile cutoff independent of assay imprecision (CV) for cardiac troponin monitoring for ruling out myocardial infarction. *Clin Chem* 2005;51:2198–200.
11. Panteghini M, Pagani F, Yeo KT, et al., on behalf of the Committee on Standardization of Markers of Cardiac Damage of the IFCC. Evaluation of imprecision for cardiac troponin assays at low-range concentrations. *Clin Chem* 2004; 50:327–32.
12. MacRae AR, Kavsak PA, Lustig V, et al. Assessing the requirement for the six-hour interval between specimens in the American Heart Association classification of myocardial infarction in epidemiology and clinical research studies. *Clin Chem* 2006;52:812–8.
13. Apple FS, Murakami MM. Cardiac troponin and creatine kinase MB monitoring during in-hospital myocardial reinfarction. *Clin Chem* 2005;51:460–3.
14. Westgard JO, Klee GG. Quality management. In: Burtis CA, Ashwood ER, Bruns DE, editors. *Tietz Textbook of Clinical Chemistry and Molecular Diagnostics*. 4th edition. St Louis, MO: Elsevier Saunders, 2006;498–9.
15. Antman EM, Anbe DT, Armstrong PW, et al. ACC/AHA guidelines for the management of patients with ST-elevation myocardial infarction: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee to Revise

- the 1999 Guidelines for the Management of Patients with Acute Myocardial Infarction) 2004. *J Am Coll Cardiol* 2004;44:671-719.
16. Braunwald E, Antman EM, Beasley JW, et al. ACC/AHA 2002 guideline update for the management of patients with unstable angina and non-ST-segment elevation myocardial infarction: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee on the Management of Patients with Unstable Angina) 2002. *J Am Coll Cardiol* 2002;40:1366-74.
17. Zimetbaum PJ, Josephson ME. Use of the electrocardiogram in acute myocardial infarction. *N Engl J Med* 2003;348:933-40.
18. Wang K, Asinger RW, Marriott HJ. ST-segment elevation in conditions other than acute myocardial infarction. *N Engl J Med* 2003;349:2128-35.
19. Holland RP, Brooks H. Precordial and epicardial surface potentials during myocardial ischemia in the pig. A theoretical and experimental analysis of the TQ and ST segments. *Circ Res* 1975;37:471-80.
20. Richeson JF, Akiyama T, Schenk E. A solid angle analysis of the epicardial ischemic TQ-ST deflection in the pig. A theoretical and experimental study. *Circ Res* 1978;43:879-88.
21. Ekmekci A, Toyoshima H, Kwoczynski JK, Nagaya T, Prinzmetal M. Angina pectoris V. Giant R wave and receding S wave in myocardial ischemia and certain non-ischemic conditions. *Am J Cardiol* 1961;7:521-32.
22. Matetzky S, Barbash GI, Rabinowitz B, et al. Q-wave and non-Q-wave myocardial infarction after thrombolysis. *J Am Coll Cardiol* 1995;26:1445-51.
23. McFarlane PW. Age, sex, and the ST amplitude in health and disease. *J Electrocardiol* 2001;34:S35-41.
24. Kligfield P, Gettes LS, Bailey JJ, et al. Recommendations for the standardization and interpretation of the electrocardiogram. Part I: the electrocardiogram and its technology. A scientific statement from the American Heart Association Electro-cardiography and Arrhythmias Committee, Council on Clinical Cardiology; the American College of Cardiology Foundation; and the Heart Rhythm Society. *J Am Coll Cardiol* 2007;49:1109-27.
25. Zimetbaum PJ, Krishnan S, Gold A, Carrozza JP II, Josephson ME. Usefulness of ST-segment elevation in lead III exceeding that of lead II for identifying the location of the totally occluded coronary artery in inferior wall myocardial infarction. *Am J Cardiol* 1998;81:918-9.
26. Engelen DJ, Gorgels AP, Cheriex EC, et al. Value of the electrocardiogram in localizing the occlusion site in the left anterior descending coronary artery in acute anterior myocardial infarction. *J Am Coll Cardiol* 1999;34:389-95.
27. Matetzky S, Freimark D, Feinberg MS, et al. Acute myocardial infarction with isolated ST segment elevation in posterior chest leads V₇-V₉. Hidden ST-segment elevations revealing acute posterior infarction. *J Am Coll Cardiol* 1999;34:748-53.
28. Agarwal JB, Khaw K, Aurignac F, LoCurto A. Importance of posterior chest leads in patients with suspected myocardial infarction, but non-diagnostic 12-lead electrocardiogram. *Am J Cardiol* 1999;83:323-6.
29. Bayés de Luna A, Wagner G, Birnbaum Y, et al. A new terminology for the left ventricular walls and for the location of myocardial infarcts that present Q wave based on the standard of cardiac magnetic resonance imaging. A statement for healthcare professionals from a Committee appointed by the International Society for Holter and Noninvasive Electrocardiography. *Circulation* 2006;114:1755-60.
30. Lopez-Sendon J, Coma-Canella I, Alcasena S, Seoane J, Gamallo C. Electrocardiographic findings in acute right ventricular infarction: sensitivity and specificity of electrocardiographic alterations in right precordial leads V₄R, V₃R, V₁, V₂ and V₃. *J Am Coll Cardiol* 1985;6:1273-9.
31. Sgarbossa EB, Pinsky SL, Barbagelata A, et al. Electrocardiographic diagnosis of evolving acute myocardial infarction in the presence of left bundle branch block. *N Engl J Med* 1996;334:481-7.
32. Wong C-K, French JK, Aylward PEG, et al. Patients with prolonged ischemic chest pain and presumed-new left bundle branch block have heterogeneous outcomes depending on the presence of ST-segment changes. *J Am Coll Cardiol* 2005;46:29-38.
33. Savage RM, Wagner GS, Ideker RE, Podolsky SA, Hackel DB. Correlation of postmortem anatomic findings with electrocardiographic changes in patients with myocardial infarction: retrospective study of patients with typical anterior and posterior infarcts. *Circulation* 1977;55:279-85.
34. Horan LG, Flowers NC, Johnson JC. Significance of the diagnostic Q wave of myocardial infarction. *Circulation* 1971;43:428-36.
35. Pahlm US, Chaitman BR, Rautaharju PM, Selvester RH, Wagner GS. Comparison of the various electrocardiographic scoring codes for estimating anatomically documented sizes of single and multiple infarcts of the left ventricle. *Am J Cardiol* 1998;81:809-15.
36. Rautaharju PM, Park LP, Chaitman BR, Rautaharju F, Zhang Z-M. The Novacode criteria for classification of ECG abnormalities and their clinically significant progression and regression. *J Electrocardiol* 1998;31:157-87.
37. Porela P, Helenius H, Pulkki K, Voipio-Pulkki LM. Epidemiological classification of acute myocardial infarction: time for a change? *Eur Heart J* 1999;20:1459-64.
38. Yokoyama Y, Chaitman BR, Hardison RM, et al. Association between new ECG abnormalities after coronary revascularization and five year cardiac mortality in BARI randomized and registry patients. *Am J Cardiol* 2000;86:819-24.
39. Korosoglou G, Labadze N, Hansen A, et al. Usefulness of real-time myocardial perfusion imaging in the evaluation of patients with first time chest pain. *Am J Cardiol* 2004;94:1225-31.
40. Dakik HA, Howell JF, Lawrie GM, et al. Assessment of myocardial viability with 99mTc-sestamibi tomography before coronary bypass graft surgery: correlation with histopathology and postoperative improvement in cardiac function. *Circulation* 1997;96:2892-8.
41. Medrano R, Lowry RW, Young JB, et al. Assessment of myocardial viability with 99mTc-sestamibi in patients undergoing cardiac transplantation: a scintigraphic/pathological study. *Circulation* 1996;94:1010-7.
42. Klein C, Nekolla SG, Bengel FM, et al. Assessment of myocardial viability with contrast-enhanced magnetic resonance imaging: comparison with positron emission tomography. *Circulation* 2002;105:162-7.
43. Wagner A, Mahrholdt H, Holly TA, et al. Contrast-enhanced MRI and routine single photon emission computed tomography (SPECT) perfusion imaging for detection of subendocardial myocardial infarcts: an imaging study. *Lancet* 2003;361:374-9.
44. Wackers FJ, Berger HJ, Johnstone DE, et al. Multiple gated cardiac blood pool imaging for left ventricular ejection fraction: validation of the technique and assessment of variability. *Am J Cardiol* 1979;43:1159-66.
45. Mahmarian JJ, Moye L, Verani MS, Eaton T, Francis M, Pratt CM. Criteria for the accurate interpretation of changes in left ventricular ejection fraction and cardiac volumes as assessed by rest and exercise gated radionuclide angiography. *J Am Coll Cardiol* 1991;18:112-9.
46. Lima JAC. Myocardial viability assessment by contrast-enhanced magnetic resonance imaging. *J Am Coll Cardiol* 2003;42:902-4.
47. Isbell DC, Kramer CM. Cardiovascular magnetic resonance: structure, function, perfusion, and viability. *J Nucl Cardiol* 2005;12:324-36.
48. Gosalia A, Haramati LB, Sheth MP, Spindola-Franco H. CT detection of acute myocardial infarction. *Am J Roentgenol* 2004;182:1563-6.
49. Mahnken AH, Koos R, Katoh M, et al. Assessment of myocardial viability in reperfused acute myocardial infarction using 16-slice computed tomography in comparison to magnetic resonance imaging. *J Am Coll Cardiol* 2005;45:2042-7.
50. Buda AJ. The role of echocardiography in the evaluation of mechanical complications of acute myocardial infarction. *Circulation* 1991;84 Suppl I:I109-21.
51. Peels C, Visser CA, Kupper AJ, Visser FC, Roos JP. Usefulness of two-dimensional echocardiography for immediate detection of myocardial ischemia in the emergency room. *Am J Cardiol* 1990;65:687-91.
52. Sabia P, Abbott RD, Afrookteh A, Keller MW, Touchstone DA, Kaul S. Importance to two-dimensional echocardiographic assessment of left ventricular systolic function in patients presenting to the emergency room with cardiac-related symptoms. *Circulation* 1991;84:1615-24.
53. Saeian K, Rhyne TL, Sagar KB. Ultrasonic tissue characterization for diagnosis of acute myocardial infarction in the coronary care unit. *Am J Cardiol* 1994;74:1211-5.

54. Tatum JL, Jesse RL, Kontos MC, et al. Comprehensive strategy for the evaluation and triage of the chest pain patients. *Ann Emerg Med* 1997;29:116-25.
55. Udelsion JE, Beshansky JR, Ballin DS, et al. Myocardial perfusion imaging for evaluation and triage of patients with suspected acute cardiac ischemia: a randomized controlled trial. *JAMA* 2002;288:2693-700.
56. Stowers SA, Eisenstein EL, Wackers FJT, et al. An economic analysis of an aggressive diagnostic strategy with single photon emission computed tomography myocardial perfusion imaging and early exercise stress testing in emergency department patients who present with chest pain but non-diagnostic electrocardiograms: results from a randomized trial. *Ann Emerg Med* 2000;35:17-25.
57. Kim RJ, Fieno DS, Parrish TB, et al. Relationship of MRI delayed contrast enhancement to irreversible necrosis, infarct age, and contractile function. *Circulation* 1999;100:1992-2002.
58. Harris BM, Nageh T, Marsden JT, Thomas MR, Sherwood RA. Comparison of cardiac troponin T and CK-MB for the detection of minor myocardial damage during interventional cardiac procedures. *Ann Clin Biochem* 2000;37:764-9.
59. Januzzi JL, Lewandrowski K, MacGillivray TE, et al. A comparison of cardiac troponin T and creatine kinase-MB for patient evaluation after cardiac surgery. *J Am Coll Cardiol* 2002;39:1518-23.
60. Holmvang L, Jurlander B, Rasmussen C, Thiis JJ, Grande P, Clemmensen P. Use of biochemical markers of infarction for diagnosing perioperative myocardial infarction and early graft occlusion after coronary artery bypass surgery. *Chest* 2002;121:103-11.
61. Miller WL, Garratt KN, Burritt MF, Reeder GS, Jaffe AS. Timing of peak troponin T and creatine kinase-MB elevations after percutaneous coronary intervention. *Chest* 2004;25:275-80.
62. Anderson KM, Califf RM, Stone GW, et al. Long-term mortality benefit with abciximab in patients undergoing percutaneous coronary intervention. *J Am Coll Cardiol* 2001;37:2059-65.
63. Akkerhuis KM, Alexander JH, Tardiff BE, et al. Minor myocardial damage and prognosis: are spontaneous and percutaneous coronary intervention-related events different? *Circulation* 2002;105:554-6.
64. Saadeddin SM, Habbab MA, Sobki SH, Ferns GA. Minor myocardial injury after elective uncomplicated successful PTCA with or without stenting: detection by cardiac troponins. *Catheter Cardiovasc Interv* 2001;53:188-92.
65. Ricciardi MJ, Wu E, Davidson CJ, et al. Visualization of discrete microinfarction after percutaneous coronary intervention associated with mild creatine kinase-MB elevation. *Circulation* 2001;103:2780-3.
66. Ricciardi MJ, Davidson CJ, Gubermikoff G, et al. Troponin I elevation and cardiac events after percutaneous coronary intervention. *Am Heart J* 2003;145:522-8.
67. Herrman J. Peri-procedural myocardial injury: 2005 update. *Eur Heart J* 2005;26:2493-519.
68. Selvanayagam JB, Porto I, Channon K, et al. Troponin elevation after percutaneous coronary intervention directly represents the extent of irreversible myocardial injury: insights from cardiovascular magnetic resonance imaging. *Circulation* 2005;111:1027-32.
69. Benoit MO, Paris M, Silleran J, Fiemeyer A, Moatti N. Cardiac troponin I: its contribution to the diagnosis of perioperative myocardial infarction and various complications of cardiac surgery. *Crit Care Med* 2001;29:1880-6.
70. Kovacevic R, Majkic-Singh N, Ignjatovic S, et al. Troponin T levels in detection of perioperative myocardial infarction after coronary artery bypass surgery. *Clin Lab* 2004;50:437-45.
71. Noora J, Ricci C, Hastings D, Hills S, Cybulsky I. Determination of troponin I release after CABG surgery. *J Card Surg* 2005;20:129-35.
72. Selvanayagam JB, Pigott D, Balacumaraswami L, Petersen SE, Neubauer S, Taggart DP. Relationship of irreversible myocardial injury to troponin I and creatine kinase-MB elevation after coronary artery bypass surgery: insights from cardiovascular magnetic resonance imaging. *J Am Coll Cardiol* 2005;45:629-31.
73. Costa MA, Carere RG, Lichtenstein SV, et al. Incidence, predictors, and significance of abnormal cardiac enzyme rise in patients treated with bypass surgery in the Arterial Revascularization Therapies Study (ARTS). *Circulation* 2001;104:2689-93.
74. Ioannidis JPA, Karvouni E, Katritsis DG. Mortality risk conferred by small elevations of creatine kinase-MB isoenzyme after percutaneous coronary intervention. *J Am Coll Cardiol* 2003;42:1406-11.
75. Coar BL, Hillis GS, Gibson PH, et al. Relationship between postoperative cardiac troponin I levels and outcome of cardiac surgery. *Circulation* 2006;114:1468-75.
76. Gustavsson CG, Hansen O, Frennby B. Troponin must be measured before and after PCI to diagnose procedure-related myocardial injury. *Scand Cardiovasc J* 2004;38:75-9.
77. Miller WL, Garratt KN, Burritt MF, Lennon RJ, Reeder GS, Jaffe AS. Baseline troponin level: key to understanding the importance of post-PCI troponin elevations. *Eur Heart J* 2006;27:1061-9.
78. Klatte K, Chaitman BR, Theroux P, et al. Increased mortality after coronary artery bypass graft surgery is associated with increased levels of postoperative creatine kinase-myocardial band isoenzyme release. *J Am Coll Cardiol* 2001;38:1070-7.
79. Brener SJ, Lytle BW, Schneider JP, Ellis SG, Topol EJ. Association between CK-MB elevation after percutaneous or surgical revascularization and three-year mortality. *J Am Coll Cardiol* 2002;40:1961-7.
80. Chew DP, Bhatt DL, Lincoff AM, Wolski K, Topol EJ. Clinical end point definitions after percutaneous coronary intervention and their relationship to late mortality: an assessment by attributable risk. *Heart* 2006;92:945-50.
81. Panteghini M, Cuccia C, Bonetti G, Giubbini R, Pagani F, Bonini E. Single-point cardiac troponin T at coronary care unit discharge after myocardial infarction correlates with infarct size and ejection fraction. *Clin Chem* 2002;48:1432-6.
82. Licka M, Zimmermann R, Zehelein J, Dengler TJ, Katus HA, Kubler W. Troponin T concentrations 72 h after myocardial infarction as a serological estimate of infarct size. *Heart* 2002;87:520-4.
83. Gallegos RP, Swingen C, Xu XJ, et al. Infarct extent by MRI correlates with peak serum troponin level in the canine model. *J Surg Res* 2004;120:266-71.
84. Meier MA, Al-Badr WH, Cooper JV, et al. The new definition of myocardial infarction: diagnostic and prognostic implications in patients with acute coronary syndromes. *Arch Intern Med* 2002;162:1585-9.
85. Kontos MC, Fritz LM, Anderson FP, Tatum JL, Ornato JP, Jesse RL. Impact of the troponin standard on the prevalence of acute myocardial infarction. *Am Heart J* 2003;146:446-52.
86. Salomaa V, Koukunen H, Ketonen M, et al. A new definition for myocardial infarction: what difference does it make? *Eur Heart J* 2005;27:1719-25.
87. Roger VL, Killian JM, Weston SA, et al. Redefinition of myocardial infarction: prospective evaluation in the community. *Circulation* 2006;114:790-7.

APPENDIX 1. RELATIONSHIPS WITH INDUSTRY—JOINT ESC/ACCF/ACA/WHF TASK FORCE FOR THE REDEFINITION OF MYOCARDIAL INFARCTION

Name	Research Contracts (Such as Grants, etc.)	Consulting/Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Writers						
Joseph S. Alpert	None	• Exeter Continuing Medical Education Company • Novartis • Sanofi-Aventis	None	None	None	None
Elliott M. Antman	• Eli Lilly (via TIMI Study Group) • Sanofi-Aventis (via TIMI Study Group)	• Sanofi-Aventis	None	None	None	None
Fred S. Apple	• Abbott • Bayer • Beckman • BioMerieux • Biosite • Clinical Diagnostics • DPC • Medical Knowledge Institute • Ortho • Roche	• Abbott • Clinical Diagnostics • Ortho • Sensera	None	None	None	None
Paul W. Armstrong	• Boehringer Ingelheim • Hoffman LaRoche Canada • Procter & Gamble/Alexion • Sanofi-Aventis • Schering-Plough Research Institute	• Abbott Laboratories • Medisure • TaregGen	None	None	None	None
Jean-Pierre Bassand	None	None	None	None	None	None
Jeroen J. Bax	• Bristol-Myers Squibb • General Electric • Guidant • Medtronic	None	None	None	None	None
George A. Beller	None	None	None	None	None	None
Robert Bonow	None	None	None	None	None	None
Bernard Chaitman	• CV Therapeutics • Pfizer	• CV Therapeutics • Merck • Pfizer • Sanofi-Aventis	None	None	None	None
Peter M. Clemmensen	• Medtronic	• Medtronic	None	None	None	None
Mikael Dellborg	None	None	None	None	None	None
T. Bruce Ferguson, Jr.	None	None	None	None	None	None
Francisco Fernández-Avilés	None	None	None			
Keith A. Fox	• Bristol-Myers Squibb • GlaxoSmithKline • Sanofi	• Bristol-Myers Squibb • GlaxoSmithKline • Sanofi	None	None	None	None

APPENDIX 1. CONTINUED

Name	Research Contracts (Such as Grants, etc.)	Consulting/Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Kim M. Fox	• Servier Labs	• Servier Labs	None	None	None	None
Marcello Galvani	None	None	None	None	None	None
Enrique P. Gurfinkel	• Sanofi-Aventis • Schering-Plough	• Sanofi-Aventis	None	None	None	None
Christian W. Hamm	• Abbott • Roche	• Abbott	None	None	None	None
Hanoch Hod	None	None	None	None	None	None
Allan S. Jaffe	• Dade-Behring	• Bayer • Beckman-Coulter • Dade-Behring • GlaxoSmithKline • Ortho Diagnostics • Pfizer	None	None	None	None
Hugo A. Katus	None	None	None	None	None	• I have invented the troponin T assay and hold a patent jointly with Roche Diagnostics
Jose-Luis Lopez-Sendon	• Amgen • AstraZeneca • Bayer • Bristol-Myers Squibb • CV Therapeutics • Eli Lilly • Merck • Novartis • Schering-Plough • Servier • Solvay	• Medtronic	None	None	None	None
Shanthi Mendis	None	None	None	None	None	None
L. Kristin Newby	• BG Medicine • Bristol-Myers Squibb-Sanofi • Inverness Medical, Inc. • Roche Diagnostics • Schering-Plough	• Adolor • Atherogenics • Biosite • Johnson & Johnson/Scios • Procter & Gamble • Tethys	None	None	None	None
E. Magnus Ohman	• Berlex* • Bristol-Myers Squibb* • Eli Lilly* • Millennium Pharmaceuticals* • Sanofi-Aventis* • Schering-Plough*	• Abiomed • Dataxcope • Invoise* • Liposcience • Response Biomedical • Savacor* • The Medicines Company		• Invoise* • Medtronic* • Savacor*		Speaker's bureau: • CV Therapeutics* • Schering-Plough*
Prem Pais	• AstraZeneca • Boehringer Ingelheim • Cadila Limited	None	None	None	None	None
Alexander N. Parkhomenko	• GlaxoSmithKline • Sanofi-Aventis • Servier	• GlaxoSmithKline	None	None	None	None

APPENDIX 1. CONTINUED

Name	Research Contracts (Such as Grants, etc.)	Consulting/Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Philip A. Poole- Wilson	• Roche	None	None	None	None	• Bayer • Menarini • Merck Sharp & Dohme
Pekka Porela	None	None	None	None	None	None
Silvia G. Priori	• Guidant • Medtronic	• Lundbeck	None	None	None	Speaker's bureau: • Guidant • Medtronic
Jan Ravkilde	• Abbott Diagnostics • Dade-Behring Diagnostics • Roche Diagnostics	None	None	None	None	None
Maarten L. Simoons	• Servier	• Servier	None	None	None	None
Philippe Gabriel Steg	• Sanofi-Aventis	• Nycomed • Sanofi-Aventis • Servier • Takeda • The Medicines Company	None	None	None	Speaker's bureau: • Boehringer- Ingelheim • Bristol-Myers Squibb • GlaxoSmithKline • Nycomed • Sanofi-Aventis • Servier • ZLB-Behring
Michal Tendera	• Amgen • Bayer • Servier	• Amgen • Bayer • Servier	None	None	None	None
Kristian Thygesen	None	None	None	None	None	None
Richard Underwood	None	• CV Therapeutics • GE Healthcare	None	None	None	None
Barry F. Uretsky	None	None	None	None	None	None
Ernst E. Van Der Wall	None	None	None	None	None	None
Liisa-Maria Voipio-Pulkki	None	None	None	None	None	None
Lars C. Wallentin	• AstraZeneca • Bristol-Myers Squibb • Boehringer Ingelheim • Eli Lilly • GlaxoSmithKline • Sanofi • Schering-Plough	None	None	None	None	None

APPENDIX 1. CONTINUED

Name	Research Contracts (Such as Grants, etc.)	Consulting/Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Harvey D. White	• Alexion • Eli Lilly • Fournier Laboratories • GlaxoSmithKline • Janssen Cilag • Johnson & Johnson • Merck Sharp & Dohme • National Institutes of Health • Neuren Pharmaceuticals • Pfizer • Procter & Gamble • Roche • Sanofi-Aventis • Schering-Plough • The Medicines Company	• GlaxoSmithKline • Medigene • Neuren Pharmaceuticals • The Medicines Company				Speaker's honorarium: • Sanofi-Aventis • The Medicines Company
William Wijns	• Abbott • Biotronic • Biotronik • Conor Medsystems • Cordis • Cordis, a Johnson & Johnson Company • GlaxoSmithKline • Medtronic • Terumo • Therabel Pharma					
David O. Williams	• Abbott Vascular • Cordis, a Johnson & Johnson Company	• Abbott Vascular • Cordis, a Johnson & Johnson Company	None	None	None	None
Jun-Ren Zhu	None	None	None	None	None	None

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*Indicates significant relationship (More than \$10,000)

APPENDIX 2. RELATIONSHIPS WITH INDUSTRY—EXTERNAL PEER REVIEW FOR THE EXPERT CONSENSUS DOCUMENT ON THE UNIVERSAL DEFINITION OF MYOCARDIAL INFARCTION

Name	Research Contracts (Such as Grants, etc.)	Consulting/Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Nawwar Al-Attar	None	None	None	None	None	None
Sorin Brenner	None	None	None	None	None	None
Darek Dudek	• Balton • Biotronic • Boehringer Ingelheim • Eli Lilly • Eurocor • Nycomed • Pfizer • Sanofi • Schering • Volcano	• Abbott • Boston • Cordis • Invatec • Medtronic				
W. Brian Gibling	• Abbott POC • i-STAT	None	None	None	None	Education grants to EMCREG- International: • Biosite • Sanofi-Aventis • Schering-Plough
David Goff	None	• GlaxoSmithKline	None	None	None	None
Robert Harrington*	• Abbott Vasc-Devices • Acorn Cardiovascular • Actelion • Acusphere, Inc. Advanced CV Syst • Advanced Stent Technologies, Inc. • Agilent • Ajinomoto • Alexion† • Alsus • Allergan • Amgen • Amylin Pharmaceuticals, Inc. • Anadys • ANGES MG, Inc. • Arginox Pharmaceuticals† • Angiometrx, Inc. • Ark Therapeutics, Ltd. • Astellas Pharma US, Inc. • Astra Hassle • AstraZeneca† • Atritech • Aventis† • Avion Flumist • Baxter • Bayer AG • Bayer Corp. • Berlex • Biocompatibles, Ltd. • Biogen • Bioheart • Biomarin • Biosense Webster, Inc. • Biosite • Bio Technology General • Biotronik • Boehringer Ingelheim	• Bristol-Myers Squibb	None	None	None	Educational activities generate activity for Duke Univ.: • Merck/Schering- Plough Venture* • Portola Pharmaceutical* • Sanofi-Aventis • Schering-Plough

APPENDIX 2. CONTINUED

Name	Research Contracts (Such as Grants, etc.)	Consulting/ Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Robert Harrington* (continued)	• Boston MedTech Advisors • Boston Scientific • Bracco Diagnostics • Bristol-Myers Squibb† • CanAm Bioresearch† • Cardiac Science, Inc. • Cardio Thoracic Systems • CardioDynamics International • CardioKinetix, Inc. • Celsion Corp. • Centocor • Chugai Pharmaceuticals • Cierra, Inc. • Coley Pharma Group • Conor Medsystems, Inc. • Corautus Genetics, Inc. • Cordis • Corgentech • Corvalent Group • Critical Therapeutics, Inc. • CryoVascular System, Inc. • Cubist Pharmaceuticals • CV Therapeutics, Inc. • Cytokinetics, Inc. • Dade Behring • Daiichi • Decide Genetics • Dyax • Echosens, Inc. • Elipse Surg Tech • Edwards Lifesciences • Eisai • Enzon Pharmaceutical • EPI-Q, Inc. • Ev3, Inc. • Evalve, Inc. • First Circle Medical, Inc. • First Horizon • Flow Cardia, Inc. • Fox Hollow Pharmaceutical • Fujisawa • Genentech • General Electric Healthcare • General Electric Medical Systems • Genzyme Corporation • Getz Bros., Co., Inc. • Gilead • GlaxoSmithKline • Guidant • Guilford Pharmaceuticals • Heartscape Technologies, Inc. • Hoffman-LaRoche • Human Genome Sciences • Humana • IDB Medical • Idenix Pharmaceuticals, Inc. • Idun Pharmaceutical, Inc. • Immunex • Idenix Pharmaceutical • INFORMD, Inc. • InfraReDx • Inhibitex • INO Therapeutics, Inc.					

APPENDIX 2. CONTINUED

Name	Research Contracts (Such as Grants, etc.)	Consulting/ Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Robert Harrington* (continued)	• Inspire Pharm • Integrated Therapeutics Group • InterMune Pharmaceuticals • Inverness Medical Innovations • ISIS Pharmaceuticals • Johnson & Johnson • Jomed, Inc. • KAI Pharmaceuticals† • Kerberos Proximal, Inc. • King Pharmaceuticals • Kuhera Chemical Co. • Lilly • Lumen Biomedical • Medacoustics • Medisure† • Medi-Flex, Inc. • Med Immune • Medtronic, Inc. • Merck • Merck/Schering-Plough Venture† • MicroMed Tech, Inc. • Microphage, Inc. • Millenium Pharmaceutical† • Mitsubishi • Momenta Pharmaceuticals • Mycosol, Inc. • Myogen • NABI • Neuron Pharmaceuticals • NitroMed • NovaCardia, Inc. • Novartis* • Organon International • Ortho Biotech • Osiris Therapeutics, Inc. • Otsuka America Pharmaceutical, Inc. • Pathway Medical Tech • PDL bioPharma • Pfizer • Pharamanetics • Pfizer • Portola Pharmaceutical† • Proctor and Gamble • Prometheus • Radiant • Recom Managed Systems • Regado Biosciences, Inc.† • Reliant Pharma • Roche Diagnostic Corp. • Salix Pharmaceuticals • Sanofi Pasteur • Sanofi-Aventis† • Sanofi-Synthelabo • Schering-Plough† • SciClone Pharmaceuticals • Scios • Searle • Sichel Technologies • Siemens					

APPENDIX 2. CONTINUED

Name	Research Contracts (Such as Grants, etc.)	Consulting/Advising	Employment in Industry	Stockholder of a Healthcare Company	Owner of a Healthcare Company (or Partnership)	Others
Robert Harrington* (continued)	• Social Scientific Systems, Inc. • Spectranetics • Summit • Suneis • TAP Pharmaceutical Products, Inc. • Terumo Corporation • The Medicines Company • Theravance • TherOx, Inc. • Titan Pharmaceuticals, Inc. • Valeant Pharmaceuticals • Valentis, Inc. • Vascular Solutions, Inc. • Velocimed • Veridex • Vertex Pharmaceuticals • Viasys Healthcare, Inc. • Vicuron Pharma • Wyeth • Wyeth-Ayerst • Xoma • Xsira Pharmaceuticals • XTL Biopharmaceuticals • Yamanouchi					
Glen N. Levine	None	None	None	None	None	None
Michael Lim	None	None	None	• Singapore Heart, Stroke and Cancer Centre	None	None
Marco A. Martínez-Rios	None	None	None	None	None	None
Joao Morals	None	• AstraZeneca	None	None	None	None
David Morrow	• Bayer Diagnostics • Beckman Coulter • Biosite Inc. • Dade-Behring • Roche Diagnostics	• Beckman Coulter	None	None	None	Honoraria for educational presentations: • Bayer Diagnostics
Udo Sechtem	• Bosch	• Novartis • Pfizer				
Steve Steinhilb	• Eli Lilly • The Medicines Company	• AstraZeneca • Eli Lilly • Sanofi-Aventis • The Medicines Company	None	None	None	None
Marco A. Tubaro	None	None	None	None	None	None

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Universal Definition of Myocardial Infarction

Kristian Thygesen, Joseph S. Alpert, Harvey D. White, on behalf of the Joint ESC/ACCF/AHA/WHF Task Force for the Redefinition of Myocardial Infarction
J. Am. Coll. Cardiol. 2007;50;2173-2195; originally published online Oct 19, 2007;
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